

These are the following perimeters on which our engine health/performance will depend.

Explanation of each and info about normal and abnormal scannerios:

```
**Example single telemetry packet (one engine, one timestamp):**  
```json  
{
 "timestamp": "2026-08-28T10:15:32Z",
 "engine_id": "ENG-04",
 "rpm": 4200,
 "cht_c": 178.5,
 "egt_c": 812.0,
 "oil_temp_c": 92.3,
 "oil_pressure_psi": 58.1,
 "fuel_flow_lph": 14.2,
 "vibration_g": 0.31,
 "battery_voltage": 13.8,
 "injection_timing_deg": 22.5,
 "altitude_m": 4200,
 "ambient_temp_c": 18.0
}
```
```

The 11 parameters

1. RPM (Revolutions Per Minute)

What it is: engine rotational speed. The baseline driver almost everything else scales against.

- Normal: cruise RPM typically sits in a defined band the engine is rated for (e.g., mid-range for a small aero piston engine); climb/high-power phases run higher.
- Abnormal: erratic fluctuation (surging) suggests fuel delivery or ignition problems; inability to reach commanded RPM suggests power loss.

2. CHT — Cylinder Head Temperature

What it is: temperature at the cylinder head, the single most-watched health parameter in piston aero engines.

- Normal: a defined safe band (too cold = incomplete combustion/wear; too hot = detonation risk).
- Abnormal: rapid rise = detonation, cooling failure, or over-lean mixture; one cylinder reading high vs. others = localized fault (stuck valve, bad plug).

3. EGT — Exhaust Gas Temperature

What it is: temperature of exhaust gases leaving the cylinder, used alongside CHT to infer mixture and combustion quality.

- Normal: tracks predictably with RPM/throttle and fuel-air mixture.
- Abnormal: sudden spike = lean condition or pre-ignition; sudden drop = misfire (cylinder not firing) — this is a textbook misfire signature.

4. Oil Pressure & Oil Temperature

What it is: lubrication system health — oil pressure keeps moving parts from grinding metal-on-metal; oil temp reflects both engine heat and lubrication efficiency.

- Normal: pressure stable within a defined range across RPM; temp rises gradually to a steady-state operating range.

- Abnormal: pressure dropping over time = developing leak or pump wear (your worked example in the execution plan); pressure oscillating = air in the system or failing pump; temp climbing without pressure change = cooling or viscosity breakdown.

5. Fuel Flow

What it is: rate of fuel consumption, should correlate tightly with RPM/power setting.

- Normal: predictable curve vs. RPM and altitude (thinner air at altitude changes mixture requirements).
- Abnormal: flow not matching expected value for RPM = injector fault, blockage, or fuel system leak.

6. Vibration Signatures

What it is: mechanical vibration amplitude/frequency pattern — arguably the earliest warning sign of many mechanical faults before temperature/pressure parameters shift at all.

- Normal: low, stable baseline amplitude at a given RPM.

- Abnormal: rising amplitude = bearing wear, propeller imbalance, or mount fatigue; specific frequency signatures can indicate *which* component (this is genuinely a deep field — vibration signature analysis is its own discipline in industrial predictive maintenance).

7. Battery/Alternator Health

What it is: electrical system status — critical because FADEC, sensors, and ignition often depend on stable electrical supply.

- Normal: voltage stable in expected range under load.
- Abnormal: voltage sag under load = failing alternator or battery degradation; this cascades to unreliable sensor readings elsewhere, which is an interesting complexity you could mention (electrical fault masquerading as sensor fault).

8. Injection Timing Parameters

What it is: precise timing of fuel injection relative to piston position — affects combustion efficiency and is usually FADEC-controlled.

- Normal: timing tracks a defined schedule vs. RPM/load.
- Abnormal: drift from commanded timing = actuator wear or FADEC calibration fault; directly causes misfire/combustion-instability symptoms.

How to get realistic data for all of these

You won't find DRDO's actual numbers — and you don't need to. Use this three-source approach:

1. **Public general-aviation reference ranges —**
Lycoming/Continental engine operating manuals and pilot-training materials publish normal operating ranges for CHT, EGT, oil temp/pressure for small aero piston engines. These are publicly available (owner's manuals, FAA/EASA general aviation training materials, engine monitor product documentation like JPI/Garmin engine monitors used in light aircraft). Use these as your **baseline "normal" ranges** — you're not claiming they're DRDO's exact numbers, just a credible, cited starting point.
2. **Automotive/industrial predictive-maintenance literature for fault signatures** — how a parameter *drifts* during a developing fault (the shape of decay,

not the exact numbers) is well documented in general condition-based-maintenance research, since the failure physics (bearing wear, lubrication breakdown, combustion instability) isn't aerospace-exclusive.